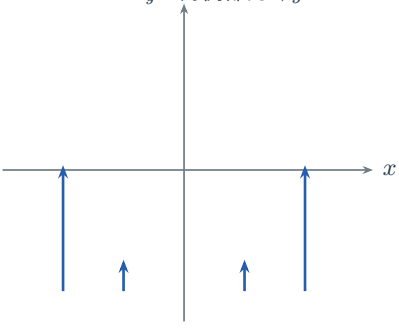
M1-11 双重旋度与拉普拉斯的符号核对

$\mathbf{A} = x^2\mathbf{e}_y$：两侧都向 $+y$



$\partial_x A_y = 2x$ 在左右两侧异号

$$\nabla \cdot \mathbf{A} = 0$$
$$\nabla \times \mathbf{A} = 2x\mathbf{e}_z$$

$$\nabla \times (\nabla \times \mathbf{A}) = (0, -2, 0)$$

$$\nabla(\nabla \cdot \mathbf{A}) - \nabla^2\mathbf{A}$$
$$= \mathbf{0} - (0, 2, 0) = (0, -2, 0)$$